

Laboratory Work #2

PART 1 - NORMALIZATION

Task 1. Which normal form is violated?

Original table:

Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

VIOLATION 1 - 1NF is broken:

Column seat_numbers stores a list: "12A, 12B"

Rule: every cell must contain exactly ONE value.

→ seat_numbers is NOT atomic → 1NF is violated.

VIOLATION 2 - 2NF is broken (after fixing 1NF):

Key becomes (booking_id, seat_number).

passenger_full_name, ticket_price depend only on booking_id,

NOT on the full key → partial dependency → 2NF is violated.

VIOLATION 3 - 3NF is broken (after fixing 2NF):

flight_number → departure_airport_name, arrival_airport_name, airline_name

Non-key gives non-key → transitive dependency → 3NF is violated.

Task 2. Step 1 - Bring to 1NF

Problem: seat_numbers = "1A, 1B" → two values in one cell.

Fix: one row per seat. New composite key: (booking_id, seat_number).

Result table - Booking_1NF:

PK: booking_id, seat_number

passenger_full_name

passenger_passport_number

flight_number

departure_airport_name, departure_city

arrival_airport_name, arrival_city

airline_name

ticket_price

Task 2. Step 2 - Bring to 2NF

Problem: passenger_*, flight_number, ticket_price depend only on booking_id (not on the full key). This is a partial dependency.

Fix: split into two tables.

Table 1 - Booking:

PK: booking_id

passenger_full_name, passenger_passport_number

flight_number, departure_airport_name, departure_city

arrival_airport_name, arrival_city, airline_name, ticket_price

Table 2 - Booking_Seat:

PK: booking_id (FK → Booking), seat_number

Task 2. Step 3 - Bring to 3NF

Problem: inside Booking, flight_number → airport names, airline_name
(transitive dependency through a non-key column).

Fix: extract Passenger, Airport, Airline, Flight.

Final 3NF tables:

Passenger (passenger_id PK, full_name, passport_number)

Airport (airport_id PK, airport_name, city)

Airline (airline_id PK, airline_name)

Flight (flight_id PK, flight_number,
departure_airport_id FK, arrival_airport_id FK, airline_id FK)

Booking (booking_id PK, passenger_id FK, flight_id FK, ticket_price)

Booking_Seat(booking_id PK FK, seat_number PK)

Task 3. Functional Dependencies

These FDs were used to justify each normalization step:

1NF → 2NF (partial dependencies - break these):

booking_id → passenger_full_name

booking_id → passenger_passport_number

booking_id → flight_number

booking_id → ticket_price

2NF → 3NF (transitive dependencies - break these):

flight_number → departure_airport_name

flight_number → arrival_airport_name

flight_number → airline_name

departure_airport_name → departure_city

arrival_airport_name → arrival_city

After 3NF - remaining (good) FDs:

booking_id, seat_number → (composite key, no partial deps)

passenger_id → full_name, passport_number

airport_id → airport_name, city

airline_id → airline_name

flight_id → flight_number, departure_airport_id, arrival_airport_id, airline_id

Task 4. Is the Flights table in 3NF?

Primary key: flight_id

Functional dependencies:

flight_id → sch_departure_time, sch_arrival_time

flight_id → act_departure_time, act_arrival_time

flight_id → departing_airport_id, arriving_airport_id

flight_id → departing_gate, arriving_gate

flight_id → airline_id, created_at, updated_at

Transitive dependency check:

departing_airport_id and airline_id are FOREIGN KEYS only.

They point to Airport and Airline tables.

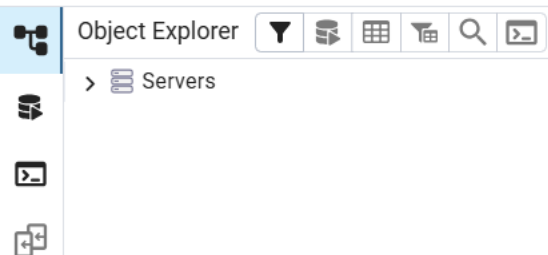
Airline name is NOT stored inside Flights.

Gates depend on the specific flight, not the airport.

→ No non-key column determines another non-key column.

Conclusion: The Flights table IS in 3NF. No change needed.

Task 5. Create 3NF tables



postgres/postgres@PostgreSQL 18*

postgres/postgres@PostgreSQL 18

Query Query History

```
1 DROP TABLE IF EXISTS Booking_Seat CASCADE;
2 DROP TABLE IF EXISTS Booking CASCADE;
3 DROP TABLE IF EXISTS Flight CASCADE;
4 DROP TABLE IF EXISTS Airport CASCADE;
5 DROP TABLE IF EXISTS Airline CASCADE;
6 DROP TABLE IF EXISTS Passenger CASCADE;
7
8 CREATE TABLE Passenger (
9     passenger_id INT PRIMARY KEY,
10    full_name     VARCHAR(100) NOT NULL,
11    passport_number VARCHAR(20) NOT NULL
12 );
13
14 CREATE TABLE Airport (
15     airport_id INT PRIMARY KEY,
16     airport_name VARCHAR(100) NOT NULL,
17     city        VARCHAR(50) NOT NULL
18 );
19
20 CREATE TABLE Airline (
```

Data Output Messages Notifications

ЗАМЕЧАНИЕ: таблица "booking_seat" не существует, пропускается
ЗАМЕЧАНИЕ: таблица "booking" не существует, пропускается
ЗАМЕЧАНИЕ: таблица "flight" не существует, пропускается
ЗАМЕЧАНИЕ: таблица "airport" не существует, пропускается
ЗАМЕЧАНИЕ: таблица "airline" не существует, пропускается
ЗАМЕЧАНИЕ: таблица "passenger" не существует, пропускается
CREATE TABLE

Query returned successfully in 80 msec.

Total rows: Query complete 00:00:00.080

CRLF

Ln 44, Col 3

Task 5. Insert data

Object Explorer

> Servers

postgres/postgres@PostgreSQL 18*

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  INSERT INTO Passenger VALUES
2  (1, 'Ivanov Ivan', 'AB1234567'),
3  (2, 'Petrova Anna', 'AC7654321'),
4  (3, 'Smagulov Timur', 'AD9988776'),
5  (4, 'Zhaksybekova Aida', 'AE1122334'),
6  (5, 'Karimov Ruslan', 'AF5566778');
7
8  INSERT INTO Airport VALUES
9  (1, 'Almaty International Airport', 'Almaty'),
10 (2, 'Astana International Airport', 'Astana'),
11 (3, 'Sheremetyevo', 'Moscow'),
12 (4, 'Istanbul Airport', 'Istanbul'),
13 (5, 'Dubai International Airport', 'Dubai');
14
15 INSERT INTO Airline VALUES
16 (1, 'Air Astana'),
17 (2, 'Aeroflot'),
18 (3, 'Turkish Airlines'),
19 (4, 'Emirates'),
20 (5, 'SCAT Airlines');
```

Data Output Messages Notifications

INSERT 0 6

Query returned successfully in 85 msec.

Total rows: Query complete 00:00:00.085

CRLF

Ln 42, Col 15

```
1 SELECT * FROM Passenger;
```

	passenger_id [PK] integer	full_name character varying (100)	passport_number character varying (20)
1	1	Ivanov Ivan	AB1234567
2	2	Petrova Anna	AC7654321
3	3	Smagulov Timur	AD9988776
4	4	Zhaksybekova Aida	AE1122334
5	5	Karimov Ruslan	AF5566778

	airport_id [PK] integer	airport_name character varying (100)	city character varying (50)
1	1	Almaty International Airp...	Almaty
2	2	Astana International Airp...	Astana
3	3	Sheremetyevo	Moscow
4	4	Istanbul Airport	Istanbul
5	5	Dubai International Airport	Dubai

	airline_id [PK] integer	airline_name character varying (100)
1	1	Air Astana
2	2	Aeroflot
3	3	Turkish Airlines
4	4	Emirates
5	5	SCAT Airlines

Object Explorer

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Query Query History

1 SELECT * FROM Flight;

Data Output Messages Notifications

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SQL

Showing rows: 1 to 5

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Page No: 1

of 1

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	flight_id [PK] integer	flight_number character varying (10)	departure_airport_id integer	arrival_airport_id integer	airline_id integer
1	1	KC101	1	2	1
2	2	SU502	1	3	2
3	3	TK712	1	4	3
4	4	EK123	2	5	4
5	5	DV234	1	2	5

Total rows: 5 Query complete 00:00:00.188

CRLF Ln 1, Col 21

	booking_id [PK] integer 	seat_number [PK] character varying (5) 
1	1001	12A
2	1002	14C
3	1003	3F
4	1004	1A
5	1004	1B
6	1005	22D

PART 2 — DDL (Airport Database)

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

Servers (1)

PostgreSQL 18

insert.sql*

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

79

booking_id

INT NOT NULL,

80

seat

VARCHAR(50) NOT NULL,

81

boarding_time

TIMESTAMP NOT NULL,

82

created_at

TIMESTAMP NOT NULL,

83

updated_at

TIMESTAMP NOT NULL

84

);

85

86

CREATE TABLE Booking_flight (

87

booking_flight_id

INT PRIMARY KEY NOT NULL,

88

booking_id

INT NOT NULL,

89

flight_id

INT NOT NULL,

90

created_at

TIMESTAMP NOT NULL,

91

updated_at

TIMESTAMP NOT NULL

92

);

93

94

CREATE TABLE Security_check (

95

security_check_id

INT PRIMARY KEY NOT NULL,

Data Output

Messages

Notifications

CREATE TABLE

Query returned successfully in 101 msec.

Total rows:

Query complete 00:00:00.101

CRLF

Ln 100, Col 3

Query returned successfully in 101 msec.

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

public

Aggregates

Collations

Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (10)

Trigger Functions

Types

Views

Subscriptions

Login/Group Roles

Tablespaces

ddl.sql x

postgres/postgres@PostgreSQL 18

No limit

QueryQuery History

```
6      created_at      TIMESTAMP NOT NULL,
7      updated_at      TIMESTAMP NOT NULL,
8      info             VARCHAR(50) NOT NULL
9  );
10
11  CREATE TABLE Airport (
12      airport_id      INT PRIMARY KEY NOT NULL,
13      airport_name     VARCHAR(50) NOT NULL,
14      country          VARCHAR(50) NOT NULL,
15      state            VARCHAR(50) NOT NULL,
16      city             VARCHAR(50) NOT NULL,
17      created_at       TIMESTAMP NOT NULL,
18      updated_at       TIMESTAMP NOT NULL
19  );
20
21  CREATE TABLE Passengers (
22      passenger_id     INT PRIMARY KEY NOT NULL,
```

Data OutputMessagesNotifications

No data output. Execute a query to get output.

Total rows: CRLF Ln 15, Col 39

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

FTS Configurations

> FTS Dictionaries

> Aa FTS Parsers

> FTS Templates

> Foreign Tables

> Functions

> Materialized Views

> Operators

> Procedures

> 1.3 Sequences

> Tables (10)

> airline_info

> airport

> baggage

> baggage_check

> boarding_pass

> booking

> Columns (8)

booking_id integer

flight_id integer

passenger_id integer

booking_platform character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

status character varying(50)

price numeric(7,2)

> Constraints

> Indexes

> RLS Policies

> Rules

> Triggers

ddl.sql x

postgres/postgres@PostgreSQL 18

No limit

QueryQuery History

6

created_at

TIMESTAMP NOT NULL,

7

updated_at

TIMESTAMP NOT NULL,

8

info

VARCHAR(50) NOT NULL

9

);

10

11

CREATE TABLE Airport (

12

airport_id

INT PRIMARY KEY NOT NULL,

13

airport_name

VARCHAR(50) NOT NULL,

14

country

VARCHAR(50) NOT NULL,

15

state

VARCHAR(50) NOT NULL,

16

city

VARCHAR(50) NOT NULL,

17

created_at

TIMESTAMP NOT NULL,

18

updated_at

TIMESTAMP NOT NULL

19

);

20

21

CREATE TABLE Passengers (

22

passenger_id

INT PRIMARY KEY NOT NULL,

Data Output

Messages

Notifications

No data output. Execute a query to get output.

Total rows:

CRLF

Ln 15, Col 39

Object Explorer

- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (10)
 - airline_info
 - Columns (7)
 - airline_id integer
 - airline_code character varying(30)
 - airline_name character varying(50)
 - airline_country character varying(50)
 - created_at timestamp without time zone
 - updated_at timestamp without time zone
 - info character varying(50)
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - airport
 - baggage

ddl.sql x

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
6      created_at      TIMESTAMP NOT NULL,
7      updated_at      TIMESTAMP NOT NULL,
8      info             VARCHAR(50) NOT NULL
9  );
10
11  CREATE TABLE Airport (
12      airport_id      INT PRIMARY KEY NOT NULL,
13      airport_name     VARCHAR(50) NOT NULL,
14      country          VARCHAR(50) NOT NULL,
15      state            VARCHAR(50) NOT NULL,
16      city             VARCHAR(50) NOT NULL,
17      created_at       TIMESTAMP NOT NULL,
18      updated_at       TIMESTAMP NOT NULL
19  );
20
21  CREATE TABLE Passengers (
22      passenger_id     INT PRIMARY KEY NOT NULL,
```

Data Output Messages Notifications

i No data output. Execute a query to get output.

Total rows:

CRLF

Ln 18, Col 36

Object Explorer

Servers (1)

PostgreSQL 18

ddl.sql*

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

1

2

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4

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7

ALTER TABLE Airline_info RENAME TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

ALTER TABLE Airline DROP COLUMN info;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 80 msec.

Total rows:

Query complete 00:00:00.080

CRLF

Ln 7, Col 38

✓ Query returned successfully in 80 msec. ✕

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

public

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FTS Configurations

FTS Dictionaries

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Foreign Tables

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Procedures

Sequences

Tables (10)

airline

airport

baggage

baggage_check

boarding_pass

booking

Columns (8)

booking_id integer

flight_id integer

passenger_id integer

booking_platform character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

status character varying(50)

ticket_price numeric(7,2)

ddl.sql*

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

1

2

3

4

5

6

7

ALTER TABLE Airline_info RENAME TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

ALTER TABLE Airline DROP COLUMN info;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 80 msec.

Total rows:

Query complete 00:00:00.080

Microsoft 365 Copilot

CRLF

Ln 7, Col 38

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

sequences

tables (10)

airline

airport

aggage

aggage_check

boarding_pass

booking

booking_flight

flights

Columns (12)

flight_id integer

sch_departure_time timestamp without time zone

sch_arrival_time timestamp without time zone

departing_airport_id integer

arriving_airport_id integer

departing_gate text

arriving_gate character varying(50)

airline_id integer

act_departure_time timestamp without time zone

act_arrival_time timestamp without time zone

created_at timestamp without time zone

updated_at timestamp without time zone

Constraints

Indexes

RLS Policies

Rules

Triggers

passengers

security_check

ddl.sql* x

postgres/postgres@PostgreSQL 18

Folder

Save

Undo

Filter

No limit

Run

Run and Stop

Refresh

Format

Table

Check

Refresh

Menu

Help

Query

Query History

1

2

3

4

5

6

7

ALTER TABLE Airline_info RENAME TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

ALTER TABLE Airline DROP COLUMN info;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 80 msec.

Total rows:

Query complete 00:00:00.080

CRLF

Ln 7, Col 38

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

public

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Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

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Materialized Views

Operators

Procedures

Sequences

Tables (10)

airline

Columns (6)

airline_id integer

airline_code character varying(30)

airline_name character varying(50)

airline_country character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

Constraints

Indexes

RLS Policies

Rules

Triggers

airport

baggage

ddl.sql*

postgres/postgres@PostgreSQL 18

No limit

Query

Query History

1 ALTER TABLE Airline_info RENAME TO Airline;

2

3 ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

4

5 ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

6

7 ALTER TABLE Airline DROP COLUMN info;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 80 msec.

Total rows: Query complete 00:00:00.080

CRLF

Ln 7, Col 38

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

Matched Views

> Operators

> Procedures

> 1.3 Sequences

> Tables (10)

> airline_info

> airport

> baggage

> baggage_check

> boarding_pass

> booking

> booking_flight

> flights

> Columns (12)

flight_id integer

sch_departure_time timestamp without time zone

sch_arrival_time timestamp without time zone

departing_airport_id integer

arriving_airport_id integer

departing_gate character varying(50)

arriving_gate character varying(50)

airline_id integer

act_departure_time timestamp without time zone

act_arrival_time timestamp without time zone

created_at timestamp without time zone

updated_at timestamp without time zone

> Constraints

> Indexes

> RLS Policies

> Rules

> Triggers

ddl.sql x

postgres/postgres@PostgreSQL 18

No limit

QueryQuery History

6created_atTIMESTAMP NOT NULL,

7updated_atTIMESTAMP NOT NULL,

8infoVARCHAR(50) NOT NULL

9);

10

11CREATE TABLE Airport (

12airport_idINT PRIMARY KEY NOT NULL,

13airport_nameVARCHAR(50) NOT NULL,

14countryVARCHAR(50) NOT NULL,

15stateVARCHAR(50) NOT NULL,

16cityVARCHAR(50) NOT NULL,

17created_atTIMESTAMP NOT NULL,

18updated_atTIMESTAMP NOT NULL

19);

20

21CREATE TABLE Passengers (

22passenger_idINT PRIMARY KEY NOT NULL,

Data OutputMessagesNotifications

No data output. Execute a query to get output.

Total rows: CRLF Ln 15, Col 39

```

1 ALTER TABLE Security_check ADD CONSTRAINT fk_sec_passenger
2     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
3
4 ALTER TABLE Booking ADD CONSTRAINT fk_booking_passenger
5     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
6
7 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck_passenger
8     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
9
10 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck_booking
11     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
12
13 ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking
14     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
15
16 ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_booking
17     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

```

ALTER TABLE

Query returned successfully in 131 msec.



- > airport
- ▼ baggage
 - > Columns
 - ▼ Constraints (1)
 - 🔑 baggage_pkey
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- ▼ baggage_check
 - > Columns
 - ▼ Constraints (1)
 - 🔑 baggage_check_pkey
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- ▼ boarding_pass
 - > Columns
 - ▼ Constraints (1)
 - 🔑 boarding_pass_pkey
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- ▼ booking
 - > Columns
 - ▼ Constraints (1)
 - 🔑 booking_pkey
 - > Indexes
 - > RLS Policies



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Query Query Histo ↗

1 ALTER TABLE Airlin

2

3 ALTER TABLE Bookin

4

5 ALTER TABLE Flight

6

7 ALTER TABLE Airlin

ges Notifications ▾ ↗

Recorded time	Event	Process
---------------	-------	---------

Total rows:

CRLF

Ln 1, Col 1


```
1 ALTER TABLE Airlines
2
3 ALTER TABLE Bookings
4
5 ALTER TABLE Flights
6
7 ALTER TABLE Airlines
```

Recorded time	Event	Process
Total rows:		
CRLF	Ln 1, Col 1	

Object Explorer

- > 1.3 Sequences
- ✓ Tables (10)
 - ✓ airline
 - > Columns
 - ✓ Constraints (1)
 - airline_info_pkey
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
 - > airport
 - ✓ baggage
 - > Columns
 - ✓ Constraints (2)
 - baggage_pkey
 - fk_baggage_booking
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
 - ✓ baggage_check
 - > Columns
 - ✓ Constraints (3)
 - baggage_check_pkey
 - fk_bagcheck_booking
 - fk_bagcheck_passenger
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
 - boarding_pass



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postgres/postgres@PostgreSQL 18

Buttons: [Folder], [Save], [Edit], [Filter], [No limit], [Run], [Refresh], [Help]

Buttons: [Editor], [Table], [Query], [Schema], [DDL], [Help]

Query	Query History
5	ALTER TABLE Booking ADD CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id) REFERENCES Passenger
6	
7	ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck FOREIGN KEY (passenger_id) REFERENCES Passenger
8	
9	
10	ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck FOREIGN KEY (booking_id) REFERENCES Booking(boo
11	
12	
13	ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking FOREIGN KEY (booking_id) REFERENCES Booking(boo
14	
15	
16	ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boardin FOREIGN KEY (booking_id) REFERENCES Booking(boo
17	
18	
19	ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookfl FOREIGN KEY (booking_id) REFERENCES Booking(boo
20	
21	
22	ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookfl FOREIGN KEY (flight_id) REFERENCES Flights(flig
23	
24	
25	ALTER TABLE Flights ADD CONSTRAINT fk_flights_dep_a FOREIGN KEY (departing_airport_id) REFERENCES A
26	
27	
28	ALTER TABLE Flights ADD CONSTRAINT fk_flights_arr_a FOREIGN KEY (arriving_airport_id) REFERENCES Ai
29	
30	
31	ALTER TABLE Flights ADD CONSTRAINT fk_flights_airli FOREIGN KEY (airline_id) REFERENCES Airline(air
32	

Total rows: CRLF Ln 1, Col 1

Object Explorer

- > Triggers
- ✓ boarding_pass
 - > Columns
 - ✓ Constraints (2)
 - boarding_pass_pkey
 - fk_boarding_booking
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- ✓ booking
 - > Columns
 - ✓ Constraints (2)
 - booking_pkey
 - fk_booking_passenger
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- ✓ booking_flight
 - > Columns
 - ✓ Constraints (3)
 - booking_flight_pkey
 - fk_bookflight_booking
 - fk_bookflight_flight
 - > Indexes
 - > RLS Policies
 - > Rules
 - > Triggers
- > flights
- > passengers



ddl.sql* x postgres/postgres@PostgreSQL 18* x

postgres/postgres@PostgreSQL 18

Buttons: [Folder], [Save], [Edit], [Filter], [No limit], [Execute], [Refresh], [Help]

Query Query History

```
1 ALTER TABLE booking ADD CONSTRAINT fk_booking_passenger
2 FOREIGN KEY (passenger_id) REFERENCES Passenger
3
4
5 FOREIGN KEY (passenger_id) REFERENCES Passenger
6
7 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagchec
8 FOREIGN KEY (passenger_id) REFERENCES Passenger
9
10 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagchec
11 FOREIGN KEY (booking_id) REFERENCES Booking(boo
12
13 ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booki
14 FOREIGN KEY (booking_id) REFERENCES Booking(boo
15
16 ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boardin
17 FOREIGN KEY (booking_id) REFERENCES Booking(boo
18
19 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookfl
20 FOREIGN KEY (booking_id) REFERENCES Booking(boo
21
22 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookfl
23 FOREIGN KEY (flight_id) REFERENCES Flights(flig
24
25 ALTER TABLE Flights ADD CONSTRAINT fk_flights_dep_a
26 FOREIGN KEY (departing_airport_id) REFERENCES A
27
28 ALTER TABLE Flights ADD CONSTRAINT fk_flights_arr_a
29 FOREIGN KEY (arriving_airport_id) REFERENCES Ai
30
31 ALTER TABLE Flights ADD CONSTRAINT fk_flights_airli
32 FOREIGN KEY (airline_id) REFERENCES Airline(air
```

Total rows: CRLF Ln 1, Col 1

Object Explorer

- Constraints (3)
 - booking_flight_pkey
 - fk_bookflight_booking
 - fk_bookflight_flight

- Indexes
- RLS Policies
- Rules
- Triggers

flights

- Columns
- Constraints (4)
 - fk_flights_airline
 - fk_flights_arr_airport
 - fk_flights_dep_airport
 - flights_pkey

- Indexes
- RLS Policies
- Rules
- Triggers

passengers

security_check

- Columns
- Constraints (2)
 - fk_sec_passenger
 - security_check_pkey

- Indexes
- RLS Policies
- Rules
- Triggers

Trigger Functions

Types



ddl.sql* x postgres/postgres@PostgreSQL 18* x

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1;
2;
3;
4;
5; passenger_id) REFERENCES Passengers(passenger_id);
6;
7; bag_check ADD CONSTRAINT fk_bagcheck_passenger
8; passenger_id) REFERENCES Passengers(passenger_id);
9;
10; bag_check ADD CONSTRAINT fk_bagcheck_booking
11; booking_id) REFERENCES Booking(booking_id);
12;
13; bag ADD CONSTRAINT fk_baggage_booking
14; booking_id) REFERENCES Booking(booking_id);
15;
16; bag_pass ADD CONSTRAINT fk_boarding_booking
17; booking_id) REFERENCES Booking(booking_id);
18;
19; flight ADD CONSTRAINT fk_bookflight_booking
20; booking_id) REFERENCES Booking(booking_id);
21;
22; flight ADD CONSTRAINT fk_bookflight_flight
23; flight_id) REFERENCES Flights(flight_id);
24;
25; ADD CONSTRAINT fk_flights_dep_airport
26; departing_airport_id) REFERENCES Airport(airport_id);
27;
28; ADD CONSTRAINT fk_flights_arr_airport
29; arriving_airport_id) REFERENCES Airport(airport_id);
30;
31; ADD CONSTRAINT fk_flights_airline
32; airline_id) REFERENCES Airline(airline_id);
```

Total rows: CRLF Ln 1, Col 1